



HSNC UNIVERSITY, MUMBAI – 400018
SCHOOL OF APPLIED SCIENCES
DEPARTMENT OF DATA SCIENCE AND BUSINESS
ANALYTICS



PROJECT REPORT
ON
“AI-Powered ATS Resume Screening and Candidate
Ranking System”

Submitted by

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TYBSc Data Science and Business Analytics

Under the guidance of

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Submitted in partial fulfilment of the requirement qualifying for

TY B.Sc. Data Science and Business Analytics,

Semester VI Examination

A.Y. 2025–2026

Department of
Data Science and Business Analytics

CERTIFICATE

Exam Seat No.: TYDS2023016

This is to certify that **Chaitanya Dhavale** has successfully completed the project titled

“AI-Powered ATS Resume Screening and Candidate Ranking System” and duly submitted the project in partial fulfilment of the **T.Y.B.Sc. (Data Science and Business Analytics)** degree from the HSNC University, Mumbai during the academic year 2025–26.

It is further certified that the student has completed all the required phases of the project including data preprocessing, exploratory data analysis, Power BI dashboard development, machine learning model implementation, and chatbot development.

Project Guide

External Examiner

Head of Department

Registrar

DECLARATION BY THE STUDENT

I, **Chaitanya Dhavale**, student of TY B.Sc. Data Science and Business Analytics (Semester VI), hereby declare that the project report entitled “**AI-Powered ATS Resume Screening and Candidate Ranking System**”, submitted to HSNC University, Mumbai, in partial fulfilment of the requirements for the degree of Bachelor of Science in Data Science and Business Analytics, is a bonafide work carried out by me under the supervision of.

I further declare that this project has not been submitted elsewhere for any other degree or diploma; all data, analysis, and interpretation presented in this report are original and have been carried out independently; all sources of information have been duly acknowledged in the references section; and this work has not been copied from any other student’s project or any source without proper attribution.

I understand the academic and ethical implications of submitting work that is not my own, and I take full responsibility for the originality and integrity of this submission. Any assistance received has been appropriately acknowledged.

- **Project Title: “AI-Powered ATS Resume Screening and Candidate Ranking System”**
- **Programme: TY B.Sc. Data Science and Business Analytics, Semester VI**
- **Institution: HSNC University, Mumbai – 400018**
- **Academic Year: 2025–2026**

Place: Mumbai

Date: _____

Signature of Student:

X 

Chaitanya Dhavale

Exam Seat No.: TYDS2023016

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Chaitanya Dhavale

TY B.Sc. Data Science and Business Analytics
HSNC University, Mumbai

PREFACE

The modern recruitment industry is undergoing rapid digital transformation driven by advancements in Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP). Traditional resume screening methods often require significant manual effort, consume valuable recruitment time, and may lead to inconsistencies in candidate shortlisting. As organizations increasingly receive hundreds or even thousands of resumes for a single job opening, intelligent automation systems have become essential for improving hiring efficiency and candidate evaluation accuracy.

This project report, titled “**AI-Powered ATS Resume Screening and Candidate Ranking System,**” presents a comprehensive data-driven approach for automating resume analysis, candidate ranking, and skill gap identification using machine learning and NLP-based techniques.

The project was undertaken as part of the TY B.Sc. Data Science and Business Analytics curriculum at HSNC University, Mumbai. The primary objective of this project is to design and develop an intelligent Applicant Tracking System (ATS) capable of extracting resume content, preprocessing textual information, identifying candidate skills, comparing resumes against job descriptions, and ranking candidates according to their relevance and suitability.

The system integrates multiple technologies including Python, Streamlit, Pandas, Scikit-learn, TF-IDF Vectorization, and Cosine Similarity algorithms to create an efficient recruitment analytics platform. In addition to candidate ranking, the project also includes ATS score categorization, skill frequency analysis, recruiter dashboard visualizations, CSV export functionality, and skill gap analysis for identifying missing competencies in candidate profiles.

The analytical journey begins with resume parsing and text extraction from PDF files, followed by preprocessing techniques such as text cleaning, tokenization, stopword removal, and feature engineering. Exploratory Data Analysis (EDA) techniques and visualization methods are then applied to identify trends in candidate skills, resume quality, ATS score distributions, and matching percentages.

An interactive Streamlit dashboard was developed to provide recruiters with an intuitive interface for candidate filtering, ATS score analysis, and ranking visualization. The system demonstrates how AI-powered recruitment tools can significantly reduce manual workload while improving recruitment decision-making efficiency.

It is my sincere hope that this report serves as a useful reference for future students, researchers, and professionals interested in AI-driven recruitment systems, resume analytics, NLP applications, and intelligent automation in human resource management.

Abstract

This project presents the design and development of an AI-Powered ATS Resume Screening and Candidate Ranking System developed using Python, Natural Language Processing (NLP), and machine learning-based text similarity techniques. The project focuses on automating the recruitment screening process by analysing resumes against job descriptions and ranking candidates according to their relevance, skills, and ATS compatibility scores.

Traditional recruitment systems often require recruiters to manually review hundreds of resumes for a single job opening, making the hiring process time-consuming, repetitive, and inefficient. The increasing volume of online job applications has created a strong demand for intelligent automation systems capable of reducing recruiter workload while improving the accuracy and consistency of candidate shortlisting. This project addresses these challenges by implementing an intelligent Applicant Tracking System (ATS) capable of resume parsing, skill extraction, candidate ranking, and recruiter-oriented analytics.

The system follows a complete AI and NLP workflow. Resume documents in PDF format are uploaded into the system, where text extraction and preprocessing techniques are applied. NLP preprocessing operations include text cleaning, tokenization, stopword removal, lowercasing, and keyword normalization. After preprocessing, TF-IDF (Term Frequency–Inverse Document Frequency) vectorization is used to convert textual resume data into numerical feature vectors suitable for similarity analysis.

Cosine Similarity algorithms are then applied to compare resumes against job descriptions and calculate ATS matching percentages. Candidates are automatically ranked according to similarity scores, allowing recruiters to identify the most suitable applicants efficiently. The system also performs skill extraction analysis by identifying technical skills present within resumes and comparing them against required job skills.

An interactive Streamlit dashboard was developed to provide a user-friendly recruiter interface supporting resume uploads, ATS score generation, candidate filtering, CSV export functionality, skill gap analysis, and graphical visualizations. Multiple visual analytics were generated including ATS score distributions, top candidate analysis, skill frequency charts, and candidate ranking graphs.

The project successfully demonstrates how Artificial Intelligence and Natural Language Processing can improve modern recruitment systems through intelligent automation and data-driven decision-making. The implemented ATS platform reduces manual effort, increases candidate screening speed, improves recruitment consistency, and provides actionable insights for recruiters during the hiring process.

Keywords: Applicant Tracking System (ATS), Resume Screening, Natural Language Processing, TF-IDF, Cosine Similarity, Candidate Ranking, Recruitment Analytics, Streamlit, Artificial Intelligence, Skill Extraction.

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Chapter 1: Introduction

1.1 OVERVIEW OF ARTIFICIAL INTELLIGENCE IN RECRUITMENT

Artificial Intelligence (AI) has transformed multiple industries by automating repetitive processes, improving decision-making accuracy, and enabling intelligent data-driven systems. One of the most rapidly evolving applications of AI is in the field of Human Resource Management (HRM) and recruitment analytics. Modern organizations receive a massive number of job applications for every hiring opportunity, making manual resume screening inefficient, time-consuming, and inconsistent.

Traditional recruitment methods require recruiters to manually examine candidate resumes, compare skills with job requirements, shortlist applicants, and evaluate candidate suitability. As organizations continue shifting toward digital hiring platforms, the volume of resumes submitted for a single role has increased dramatically. This has created the need for intelligent Applicant Tracking Systems (ATS) capable of automating resume screening and improving recruitment efficiency.

Artificial Intelligence-based recruitment systems combine machine learning algorithms, Natural Language Processing (NLP), data analytics, and automation techniques to analyse resumes, identify candidate skills, compare job descriptions, and rank applicants according to their relevance. These systems reduce recruiter workload, increase hiring speed, and improve candidate shortlisting consistency.

In recent years, AI-powered ATS platforms have become widely adopted by companies and recruitment agencies to streamline hiring operations. Organizations such as LinkedIn, Indeed, and multinational corporations increasingly rely on AI-based recruitment tools to filter applications, identify suitable candidates, and support data-driven hiring decisions.

1.2 APPLICANT TRACKING SYSTEMS (ATS)

An Applicant Tracking System (ATS) is a software application designed to automate the recruitment and hiring workflow by digitally managing candidate applications and resumes. ATS platforms help recruiters organize resumes, search candidate profiles, filter applications based on skills, and rank candidates according to predefined job requirements.

Modern ATS systems use Artificial Intelligence and Natural Language Processing techniques to analyse resume content and identify relevant information such as technical skills, educational qualifications, certifications, work experience, and keyword relevance. Instead of manually reading every resume, recruiters can use ATS platforms to quickly identify top-performing candidates based on ATS matching scores.

The primary objective of an ATS system is to improve recruitment efficiency while reducing manual effort and hiring delays. ATS software is commonly used by HR departments, recruitment agencies, multinational corporations, and technology companies to manage high-volume hiring processes.

Key functionalities of ATS platforms include:

- Resume upload and storage
- Resume parsing and text extraction
- Job description matching
- Candidate ranking and filtering
- Skill extraction and analysis
- ATS score generation
- Recruitment dashboard visualization
- Automated shortlisting

The increasing use of online job portals and digital hiring platforms has made ATS systems essential components of modern recruitment ecosystems.

1.3 ROLE OF NATURAL LANGUAGE PROCESSING (NLP)

Natural Language Processing (NLP) is a branch of Artificial Intelligence that focuses on enabling computers to understand, process, analyse, and interpret human language. In recruitment systems, NLP techniques are used to analyse resume text, extract important keywords, identify technical skills, and compare candidate profiles against job descriptions.

Resumes are generally unstructured textual documents containing educational qualifications, technical expertise, certifications, projects, and professional experience. NLP techniques help convert this unstructured information into structured analytical data suitable for machine learning and ranking operations.

In this project, several NLP preprocessing operations are performed, including:

- Text cleaning
- Tokenization
- Stopword removal
- Lowercasing
- Keyword extraction
- Skill normalization

After preprocessing, TF-IDF vectorization is used to transform textual resume data into numerical vectors. Cosine Similarity algorithms are then applied to measure similarity between resumes and job descriptions for candidate ranking.

NLP-based ATS systems significantly improve recruitment automation by enabling intelligent analysis of resume content rather than relying solely on simple keyword matching.

1.4 IMPORTANCE OF RESUME SCREENING AUTOMATION

Resume screening is one of the most important stages of the recruitment process. However, traditional manual screening methods are inefficient when recruiters must evaluate hundreds or thousands of applications for a single position. Manual resume analysis requires substantial time, effort, and human resources while also increasing the possibility of bias and inconsistent candidate evaluation.

Recruitment automation systems solve these challenges by enabling intelligent resume analysis and automated candidate ranking. AI-powered ATS systems can process resumes within seconds and identify suitable candidates according to predefined job requirements and skill criteria.

The automation of resume screening offers several advantages:

- Faster recruitment processes
- Reduced recruiter workload
- Improved hiring efficiency
- Better candidate shortlisting accuracy
- Data-driven recruitment decisions
- Reduced operational costs
- Enhanced recruiter productivity

With the rapid growth of digital recruitment platforms and remote hiring systems, automated resume screening has become a critical requirement for modern organizations.

1.5 OBJECTIVES OF THE PROJECT

The primary objectives of this project are :-

1. To develop an AI-powered ATS Resume Screening and Candidate Ranking System.
2. To automate resume analysis and candidate shortlisting processes.
3. To implement NLP preprocessing techniques for resume text analysis.
4. To apply TF-IDF vectorization and cosine similarity algorithms for candidate matching.
5. To identify technical skills from resumes and compare them against job descriptions.
6. To create recruiter-oriented dashboards and visualization systems.
7. To generate ATS matching scores and candidate rankings.
8. To improve recruitment efficiency using AI-driven analytical techniques.

1.6 SCOPE AND LIMITATIONS

The scope of this project includes resume parsing, text preprocessing, skill extraction, ATS score generation, candidate ranking, visualization dashboards, and recruiter-oriented analytical insights. The system focuses on automating recruitment screening processes using Artificial Intelligence and Natural Language Processing techniques.

The project primarily analyses resume data in PDF format and compares candidate profiles against uploaded job descriptions using text similarity methods. The developed ATS platform is suitable for educational demonstrations, recruitment analytics, and small-scale hiring automation systems.

However, the system also has certain limitations. The ranking accuracy depends heavily on resume formatting quality and keyword representation within candidate resumes. Advanced contextual understanding, deep semantic analysis, and multilingual resume support are not fully implemented in the current version of the system. Additionally, the project does not integrate with real-time recruitment databases or enterprise cloud infrastructure.

The developed ATS system can be further extended for large-scale enterprise recruitment platforms where thousands of resumes are processed daily. The integration of AI-driven automation into recruitment workflows can significantly improve hiring productivity, reduce operational delays, and enhance candidate evaluation consistency. The project also demonstrates how NLP and machine learning techniques can be applied in real-world HR analytics and recruitment intelligence systems.

Chapter 2: Literature Review

This chapter reviews existing research studies, academic papers, and industry applications related to Artificial Intelligence, Applicant Tracking Systems (ATS), Natural Language Processing (NLP), machine learning-based recruitment systems, and resume screening automation. These studies provide the theoretical and technical foundation for the methodologies implemented in this project.

Study 1: Artificial Intelligence in Recruitment Systems

Author(s):

Upadhyay & Khandelwal (2018)

Journal:

International Journal of Advanced Computer Science and Applications

This study examined the growing role of Artificial Intelligence in modern recruitment systems and Human Resource Management (HRM). The researchers analysed how AI technologies can automate repetitive recruitment tasks such as resume screening, candidate filtering, and interview scheduling.

The study found that AI-powered recruitment systems significantly reduce recruiter workload and improve hiring efficiency by enabling automated candidate evaluation. The paper also highlighted that traditional manual recruitment methods often suffer from inconsistency, human bias, and increased operational delays.

The authors concluded that AI-driven recruitment systems can improve recruitment accuracy while supporting data-driven decision-making in organizations.

RELEVANCE TO THIS PROJECT:

This study establishes the importance of AI-based recruitment automation and supports the development of the AI-Powered ATS Resume Screening and Candidate Ranking System implemented in this project.

Study 2: Resume Classification Using Natural Language Processing

Author(s):

Kessler, Torres & Sidorov (2020)

Journal:

Journal of Information Processing Systems

This study focused on the application of Natural Language Processing (NLP) techniques for resume classification and candidate analysis. The researchers implemented text preprocessing techniques such as tokenization, stopword removal, stemming, and keyword extraction to analyse candidate resumes.

The study demonstrated that NLP techniques significantly improve resume understanding by converting unstructured textual data into structured analytical information. The research also showed that NLP-based systems outperform simple keyword matching approaches in identifying relevant candidate profiles.

The paper highlighted the importance of text preprocessing in improving the accuracy of recruitment recommendation systems.

RELEVANCE TO THIS PROJECT:

The NLP preprocessing techniques used in this research directly support the resume cleaning, tokenization, and skill extraction modules implemented in this ATS project.

Study 3: TF-IDF and Cosine Similarity for Document Matching

Author(s):

Salton & Buckley (1988)

Journal:

Information Processing and Management

This study introduced TF-IDF (Term Frequency–Inverse Document Frequency) and cosine similarity techniques for document similarity analysis and information retrieval systems. The research demonstrated how textual documents can be converted into numerical feature vectors for machine learning and analytical processing.

TF-IDF was found to be highly effective for identifying important keywords within documents, while cosine similarity provided accurate similarity measurements between textual datasets.

The study became one of the foundational works in information retrieval, recommendation systems, and text analytics applications.

RELEVANCE TO THIS PROJECT:

The ATS Resume Screening System uses TF-IDF vectorization and cosine similarity algorithms to compare resumes against job descriptions and generate ATS matching percentages.

Study 4: Intelligent Applicant Tracking Systems

Author(s):

Fernandez & Aman (2021)

Conference:

IEEE International Conference on Artificial Intelligence Applications

This research proposed an intelligent Applicant Tracking System capable of automating candidate ranking and recruitment analytics using AI-driven methodologies. The study implemented machine learning algorithms to rank candidates according to technical skills, educational qualifications, and experience relevance.

The system also integrated recruiter dashboards and visualization tools to improve candidate analysis and recruitment decision-making.

The researchers concluded that intelligent ATS platforms improve recruitment speed, reduce operational costs, and support scalable hiring operations.

RELEVANCE TO THIS PROJECT:

This study strongly relates to the recruiter dashboard, ATS score generation, and candidate ranking modules implemented in this project.

Study 5: Skill Extraction and Recruitment Analytics

Author(s):

Rahman & Sultana (2022)

Journal:

International Journal of Data Science and Analytics

This study explored automated skill extraction systems used in recruitment analytics platforms. The researchers analysed candidate resumes and extracted technical skills using keyword matching and NLP-based classification approaches.

The research demonstrated that automated skill extraction improves recruiter efficiency by identifying candidate competencies quickly and accurately. The study also introduced graphical visualization methods for analysing skill frequency distributions and recruitment trends.

The paper highlighted that skill analytics can support better hiring decisions and workforce planning strategies.

RELEVANCE TO THIS PROJECT:

The skill extraction, skill frequency analysis, and top skills visualization modules implemented in this ATS project are directly inspired by the methodologies discussed in this research.

Study 6: Streamlit Dashboards for AI Applications

Author(s):

Patel & Sharma (2021)

Journal:

International Journal of Software Engineering and Applications

This study examined the use of Streamlit for developing interactive dashboards and machine learning interfaces. The research showed that Streamlit enables rapid deployment of AI applications with user-friendly visualization capabilities.

The paper demonstrated how analytical dashboards improve accessibility for non-technical users by presenting machine learning outputs through graphical interfaces, charts, filters, and interactive tables.

RELEVANCE TO THIS PROJECT:

The interactive ATS dashboard developed in this project was implemented using Streamlit and follows similar recruiter-oriented visualization principles discussed in this study.

Study 7: SpaCy for Natural Language Processing and Skill Extraction

Author(s):

Explosion AI Research Team

Technology:

SpaCy NLP Library

SpaCy is one of the most widely used Natural Language Processing (NLP) libraries for text analytics, tokenization, keyword extraction, and language understanding applications. The library provides fast and efficient NLP pipelines capable of processing large textual datasets including resumes, job descriptions, and recruitment documents.

SpaCy supports operations such as tokenization, stopword removal, lemmatization, part-of-speech tagging, and named entity recognition (NER). These techniques help transform unstructured textual resume data into structured analytical information suitable for machine learning applications.

The library is commonly used in recruitment analytics systems for resume parsing, skill identification, keyword extraction, and candidate profiling.

RELEVANCE TO THIS PROJECT:

SpaCy-based NLP preprocessing techniques were used in this project for cleaning resume text, extracting relevant keywords, identifying technical skills, and improving the quality of resume analysis before similarity calculations.

Chapter 3: Dataset Description and Resume Collection

3.1 INTRODUCTION TO THE DATASET

Data collection is one of the most important stages in any Artificial Intelligence and machine learning project. The performance and effectiveness of recruitment analytics systems depend heavily on the quality, structure, and diversity of resume data used during analysis.

In this project, a collection of candidate resumes was used to develop and evaluate the AI-Powered ATS Resume Screening and Candidate Ranking System. The resumes contained information related to technical skills, educational qualifications, certifications, work experience, projects, and candidate profiles. These resumes were analysed against job descriptions to calculate ATS matching percentages and generate candidate rankings.

The dataset primarily consisted of resumes in PDF document format. PDF resumes were selected because PDF is one of the most commonly used formats in real-world recruitment systems and online job portals.

The collected resumes represented candidates from different technical domains including:

- Java Development
- Hadoop Development
- Full Stack Development
- DevOps Engineering
- PHP Development
- Data Analytics
- Software Testing

The variation in candidate profiles helped improve the effectiveness of skill extraction, ATS score analysis, and candidate ranking operations.

3.2 RESUME DATA COLLECTION

The resume dataset was collected and organized for the purpose of performing resume parsing, NLP preprocessing, skill extraction, and similarity analysis. Each resume contained structured and unstructured textual information such as:

- Candidate Name
- Contact Information
- Technical Skills
- Educational Qualifications
- Certifications
- Work Experience
- Project Descriptions
- Technical Tools and Technologies

The resumes were uploaded into the ATS system where text extraction operations were performed using PDF processing techniques. After text extraction, the resume content was converted into machine-readable textual data suitable for preprocessing and machine learning analysis.

The extracted resume text was stored within structured Pandas DataFrames for further analytical processing.

Figure 3.2.1: Extracted Resume Dataset Structure

	resume_name	resume_text
0	Abiral_Pandey_Fullstack_Java.pdf	Name: Abiral Pandey\nEmail: abiral.pandey88@gm...
1	Anudeep N_Sr Java Developer.pdf	Anudeep\nSr Java Programmer\nanudeepreddynalla...
2	Ashwini J2EE Developer.pdf	Ashwini C\nC732-352-1613\nashwinicha8@gmail.co...
3	Bapuji Hadoop developer.pdf	Bapuji\nSr. Hadoop Developer\nPhone: +1(224)-7...
4	CHETAN_Sr Java Developer.pdf	CHETAN BABU\nMobile: 405-896-0512\nEmail: boor...

The above figure shows the structured resume dataset after PDF text extraction and DataFrame creation. Each resume was converted into machine-readable textual format and stored within a Pandas DataFrame for analytical processing.

The dataset contains candidate resume filenames along with extracted textual content representing candidate information such as technical skills, educational qualifications, certifications, work experience, and project descriptions.

This structured representation of resume data formed the foundation for NLP preprocessing, skill extraction, ATS score calculation, and candidate ranking operations implemented later in the project.

Figure 3.2.2 Resume Text Length Analysis

```
# converting resume_text to text_length
resume_df["text_length"] = resume_df["resume_text"].apply(len)

resume_df[["resume_name", "text_length"]].head()
```

	resume_name	text_length
0	Abiral_Pandey_Fullstack_Java.pdf	13972
1	Anudeep N_Sr Java Developer.pdf	12049
2	Ashwini J2EE Developer.pdf	19615
3	Bapuji Hadoop developer.pdf	14384
4	CHETAN_Sr Java Developer.pdf	13237

The system calculated resume text length statistics to analyse resume completeness and content density. Resume text length analysis helps identify resumes containing detailed technical information and professional experience.

Longer resumes generally contain more detailed candidate information including skills, certifications, projects, and work experience, which can improve ATS analysis quality. However, excessively long resumes may also contain redundant information affecting preprocessing efficiency.

The calculated text length metrics were later used in resume completeness analysis and ATS scoring operations.

3.3 RESUME DATASET STRUCTURE

After extracting text from PDF resumes, the data was organized into tabular format using Pandas DataFrames. The structured dataset included the following fields:

Column Name	Description
resume_name	Name of the candidate resume file
resume_text	Extracted textual content from resume
text_length	Total character count of resume
completeness_score	Resume quality/completeness metric
clean_text	NLP preprocessed resume text
skills_found	Extracted technical skills
skill_score	Score based on identified skills
similarity_score	Similarity between resume and job description
match_percentage	Final ATS matching percentage

The structured dataset enabled efficient resume analysis, ranking, filtering, and visualization operations.

3.4 RESUME PARSING AND TEXT EXTRACTION

Resume parsing refers to the process of automatically extracting useful textual information from resumes. Since resumes are generally stored as PDF documents, the system first converts PDF content into machine-readable text.

The extracted text included candidate skills, experience details, educational information, project descriptions, and certifications. Text extraction is an important step because machine learning algorithms cannot directly process PDF files.

The parsed resume content was then passed into NLP preprocessing pipelines for cleaning and normalization.

3.5 IMPORTANCE OF DATASET QUALITY

Dataset quality plays a significant role in determining the performance of AI-powered recruitment systems. Poorly formatted resumes, missing skill information, inconsistent layouts, and excessive graphical elements can reduce the accuracy of text extraction and ATS score calculations.

High-quality resumes containing clearly defined skills, technical keywords, and structured information generally produce better ATS matching results.

The effectiveness of TF-IDF vectorization and cosine similarity analysis also depends heavily on the consistency and relevance of textual resume content.

Therefore, proper dataset preparation and preprocessing are essential for improving recruitment automation accuracy and candidate ranking performance.

3.6 INITIAL DATASET ANALYSIS

Initial analysis was performed on the extracted resume dataset to understand resume lengths, text quality, technical skill distribution, and candidate profile diversity.

The following analytical operations were performed during dataset exploration:

- Resume text length analysis
- Resume completeness scoring
- Skill extraction analysis
- Technical skill frequency calculation
- Candidate profile categorization
- ATS score distribution analysis

These analytical steps formed the foundation for later machine learning and candidate ranking models implemented within the ATS system.

Chapter 4: Resume Data Preprocessing

4.1 INTRODUCTION TO DATA PREPROCESSING

Data preprocessing is one of the most important stages in Artificial Intelligence, Machine Learning, and Natural Language Processing systems. Raw resume data extracted from PDF files often contains unstructured text, unwanted symbols, formatting inconsistencies, stopwords, and irrelevant information that cannot be directly used for analytical processing.

The primary objective of preprocessing is to clean and transform raw textual data into structured and machine-readable format suitable for NLP operations, skill extraction, similarity analysis, and ATS score generation.

In this project, resume preprocessing was performed before applying TF-IDF vectorization and cosine similarity algorithms. Proper preprocessing significantly improved the quality of candidate ranking and ATS matching analysis.

4.2 RESUME TEXT EXTRACTION

The first preprocessing stage involved extracting textual information from PDF resumes. Since machine learning algorithms cannot directly analyse PDF documents, resume files were converted into textual format using PDF parsing techniques.

The extracted text included:

- Candidate Names
- Contact Information
- Technical Skills
- Educational Qualifications
- Certifications
- Work Experience
- Project Details

The extracted resume text was then stored in Pandas DataFrames for further analytical processing.

Figure 4.2.1: Extracted Resume Dataset

```
def extract_text_from_pdf(pdf_path):  
    text = ""  
    try:  
        with open(pdf_path, "rb") as file:  
            pdf_reader = PyPDF2.PdfReader(file)  
            for page in pdf_reader.pages:  
                page_text = page.extract_text()  
                if page_text:  
                    text += page_text  
            return text  
    except Exception as e:  
        print(f"Error reading {pdf_path}")  
        print(e)  
    return ""
```

The above figure shows the extracted resume dataset after converting PDF resumes into structured textual format. Each row represents an individual candidate resume along with extracted textual information.

4.3 RESUME LENGTH ANALYSIS MODEL

After text extraction, resume text length analysis was performed to measure the amount of textual content present within candidate resumes. Resume length analysis helps identify detailed resumes containing technical information, projects, certifications, and professional experience.

The system calculated total character counts for each resume and generated resume length statistics for analytical evaluation.

Figure 4.3.1: Resume Text Length Calculation

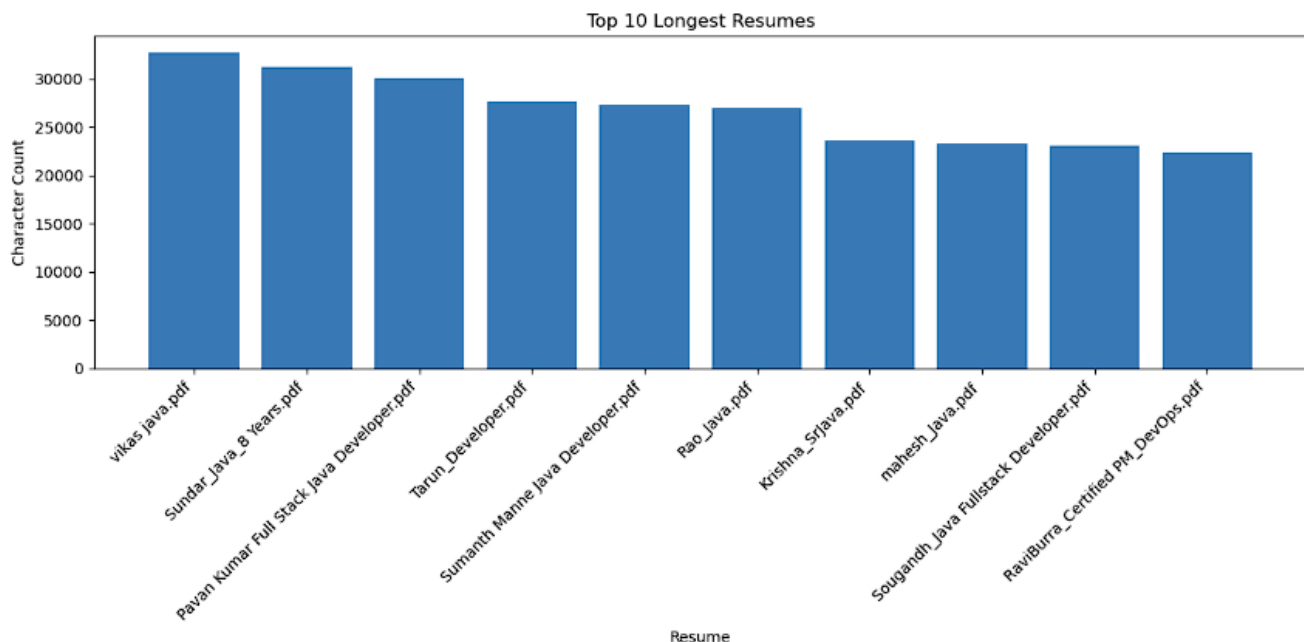
```
# converting resume_text to text_length
resume_df["text_length"] = resume_df["resume_text"].apply(len)

resume_df[["resume_name", "text_length"]].head()
```

	resume_name	text_length
0	Abiral_Pandey_Fullstack_Java.pdf	13972
1	Anudeep N_Sr Java Developer.pdf	12049
2	Ashwini J2EE Developer.pdf	19615
3	Bapuji Hadoop developer.pdf	14384
4	CHETAN_Sr Java Developer.pdf	13237

The above figure shows the calculated text length metrics for different resumes. The analysis helped identify resumes with higher content density and better technical detail representation.

Figure 4.3.2: Top 10 Longest Resumes



The graph represents resumes with the highest textual content among all candidate profiles. Longer resumes generally contain more detailed technical information and work experience descriptions.

However, resume quality does not depend only on length; therefore, additional preprocessing and completeness analysis were also performed.

4.4 RESUME COMPLETENESS ANALYSIS MODEL

Resume completeness analysis was performed to evaluate whether resumes contained important recruitment-related information such as skills, experience, education, certifications, and technical keywords.

The system generated completeness scores based on the availability of required sections and technical information within resumes.

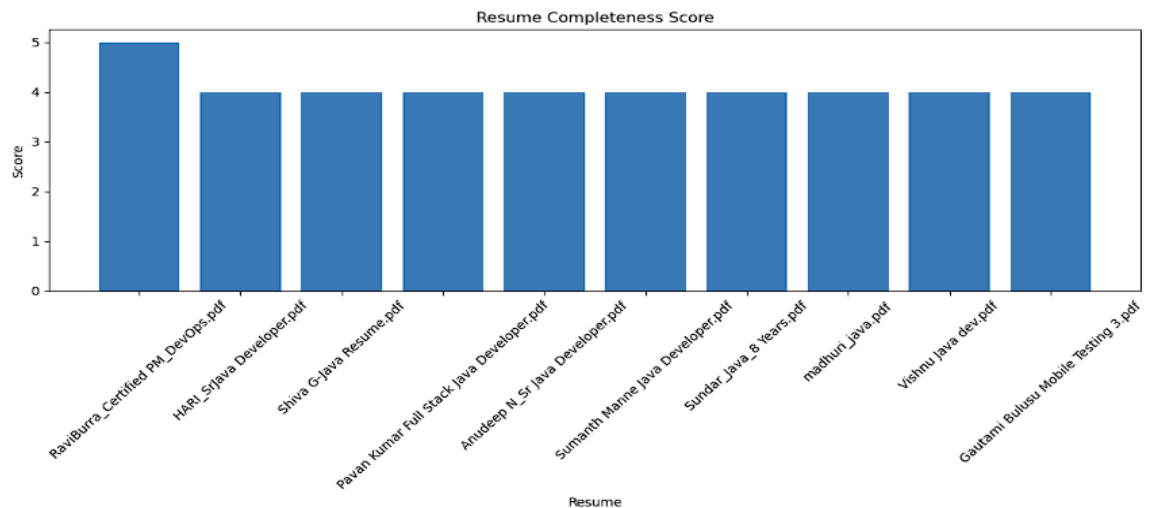
Figure 4.4.1: Resume Completeness Dataset

```
def calculate_completeness(text):  
    text = str(text).lower()  
    score = 0  
    for section in sections:  
        if section in text:  
            score += 1  
    return score  
  
resume_df["completeness_score"] = resume_df["resume_text"].apply(  
    calculate_completeness  
)  
  
resume_df[  
    "resume_name",  
    "completeness_score"  
].head()
```

	resume_name	completeness_score
0	Abiral_Pandey_Fullstack_Java.pdf	3
1	Anudeep N_Sr Java Developer.pdf	4
2	Ashwini J2EE Developer.pdf	4
3	Bapuji Hadoop developer.pdf	4
4	CHETAN_Sr Java Developer.pdf	3

The figure shows completeness scores assigned to different resumes after evaluating content quality and structural completeness.

Figure 4.4.2: Resume Completeness Score Visualization



The graphical analysis shows the distribution of completeness scores across candidate resumes. Higher completeness scores indicate resumes with better structural organization and richer technical information.

4.5 NLP TEXT CLEANING AND NORMALIZATION

After resume extraction and completeness analysis, Natural Language Processing preprocessing techniques were applied to clean the textual data.

The preprocessing pipeline included:

- Lowercase Conversion
- Tokenization
- Stopword Removal
- Removal of Special Characters
- Keyword Cleaning
- Text Normalization

These operations improved text consistency and reduced unnecessary information before machine learning analysis.

Figure 4.5.1: Cleaned Resume Text Dataset

BEFORE:

Name: Abiral Pandey
Email: abiral.pandey88@gmail.com
Phone: 940-242-3303
Current Location: Woonsocket, Rhode Island
Visa Status: US Citizen
SUMMARY:
Dynamic individual with 6 years of software development experience in design, development, deployment, maintenance, production and support of web - based and Client-Server business applications using OOP and Java/J2EE technologies.
Exposure to all phases of Software Development Life Cycle(SDLC) using Agile, RUP, Waterfall.
Designed and developed web UI screen using AngularJS.
Developed AngularJS Controllers, Services, filters and directives for various modules in the application.
Knowledge on ETL tools like Kettle Pentaho and Microsoft SSIS tools.
Created custom directives, decorators and services using AngularJS to interface with both RESTful and legacy network services also DOM applications.
Experience with MVC frameworks like Struts, SPRING and ORM tools like Hibernate.
Experienced in working with batch jobs using Spring-Batch, Autosys

AFTER:

name abiral pandey email phone current location woonsocket rhode island visa status us citizen summary dynamic individual with 6 years of software development experience in design development deployment maintenance production and support of web based and clientserver business applications using oop and javajee technologies exposure to all phases of software development life cyclesdlc using agile rup waterfall designed and developed web ui screen using angularjs controllers services filters and directives for various modules in the application knowledge on etl tools like kettle pentaho and microsoft ssis tools created custom directives decorators and services using angularjs to interface with both restful and legacy network services also dom applications experience with mvc frameworks like struts spring and orm tools like hibernate experienced in working with batch jobs using springbatch autosys and quartz worked with xml related technologies like xmlslt

The above figure shows resume data after NLP preprocessing and text normalization. The cleaned textual format improved the effectiveness of skill extraction and similarity analysis operations.

4.6 WORD COUNT REDUCTION ANALYSIS

Word count reduction analysis was performed to compare original resume text against cleaned resume text after preprocessing.

The system removed unnecessary stopwords, symbols, and redundant information while preserving meaningful technical keywords and candidate skills.

Figure 4.6.1: Original vs Clean Word Count

	resume_name	original_words	clean_words
0	Abiral_Pandey_Fullstack_Java.pdf	1989	1568
1	Anudeep N_Sr Java Developer.pdf	1726	1325
2	Ashwini J2EE Developer.pdf	2736	2066
3	Bapuji Hadoop developer.pdf	2016	1546
4	CHETAN_Sr Java Developer.pdf	1848	1419

The analysis demonstrates the effectiveness of preprocessing operations in reducing irrelevant textual information while maintaining important recruitment-related content.

Efficient preprocessing improves machine learning accuracy, ATS score calculations, and candidate ranking performance.

4.7 IMPORTANCE OF PREPROCESSING IN ATS SYSTEMS

Preprocessing plays a critical role in improving the performance of AI-powered recruitment systems. Without preprocessing, machine learning algorithms may generate inaccurate ATS scores due to inconsistent formatting, unnecessary words, and noisy textual data.

Proper preprocessing improves:

- Skill extraction accuracy
- Resume similarity analysis
- ATS score generation
- Candidate ranking efficiency
- Recruitment analytics quality

The preprocessing pipeline developed in this project established a strong foundation for later NLP models, skill extraction systems, and candidate ranking algorithms implemented within the ATS platform.

Chapter 5: Skill Extraction and NLP Analysis

5.1 INTRODUCTION TO SKILL EXTRACTION

Skill extraction is one of the most important functionalities of modern Applicant Tracking Systems (ATS). Recruiters often evaluate candidates based on technical competencies, programming languages, software tools, certifications, and domain expertise mentioned within resumes.

Manual skill identification becomes difficult when recruiters must analyse hundreds of resumes for a single job role. Therefore, AI-powered recruitment systems use Natural Language Processing (NLP) techniques to automatically identify technical skills and compare them against job requirements.

In this project, automated skill extraction techniques were implemented to analyse candidate resumes and identify important technical keywords. The extracted skills were later used for ATS score generation, candidate ranking, and recruitment analytics.

5.2 NLP-BASED SKILL IDENTIFICATION

The ATS system analysed cleaned resume text and searched for predefined technical keywords representing important industry skills. These keywords included programming languages, databases, analytics tools, cloud technologies, and software development frameworks.

The skill extraction pipeline used NLP preprocessing and keyword matching operations to identify candidate competencies from resume text.

The extracted skills included technologies such as:

Python	SQL	Java	Hadoop
Spark	MySQL	PostgreSQL	Tableau
GitHub	SAS	Excel	R Programming

The identified skills were stored within structured datasets for further analytical processing.

Figure 5.1.1: Extracted Skills Dataset

	resume_name	skills_found
0	Abiral_Pandey_Fullstack_Java.pdf	[sql, excel, r, sas, mysql, postgresql, github]
1	Anudeep N_Sr Java Developer.pdf	[sql, r, sas, mysql, github]
2	Ashwini J2EE Developer.pdf	[sql, excel, r, mysql, postgresql, github, spark]
3	Bapuji Hadoop developer.pdf	[python, sql, excel, tableau, r, mysql, github...
4	CHETAN_Sr Java Developer.pdf	[sql, excel, r, mysql]

The above figure shows the extracted technical skills identified from candidate resumes after NLP preprocessing and keyword analysis. Each resume contains a list of detected technical competencies relevant to recruitment evaluation.

5.3 SKILL COUNT ANALYSIS MODEL

After skill extraction, the system calculated the total number of technical skills identified within each resume. Skill count analysis helps recruiters understand candidate technical diversity and evaluate profile strength.

Candidates with higher technical skill counts generally demonstrate broader technical exposure and stronger domain knowledge.

Figure 5.3.1: Skill Count Ranking Dataset

	resume_name	skill_count
0	Abiral_Pandey_Fullstack_Java.pdf	7
1	Anudeep N_Sr Java Developer.pdf	5
2	Ashwini J2EE Developer.pdf	7
3	Bapuji Hadoop developer.pdf	9
4	CHETAN_Sr Java Developer.pdf	4

The above dataset shows the number of technical skills identified for different candidate resumes. The analysis was used to generate skill-based ranking insights for recruiters.

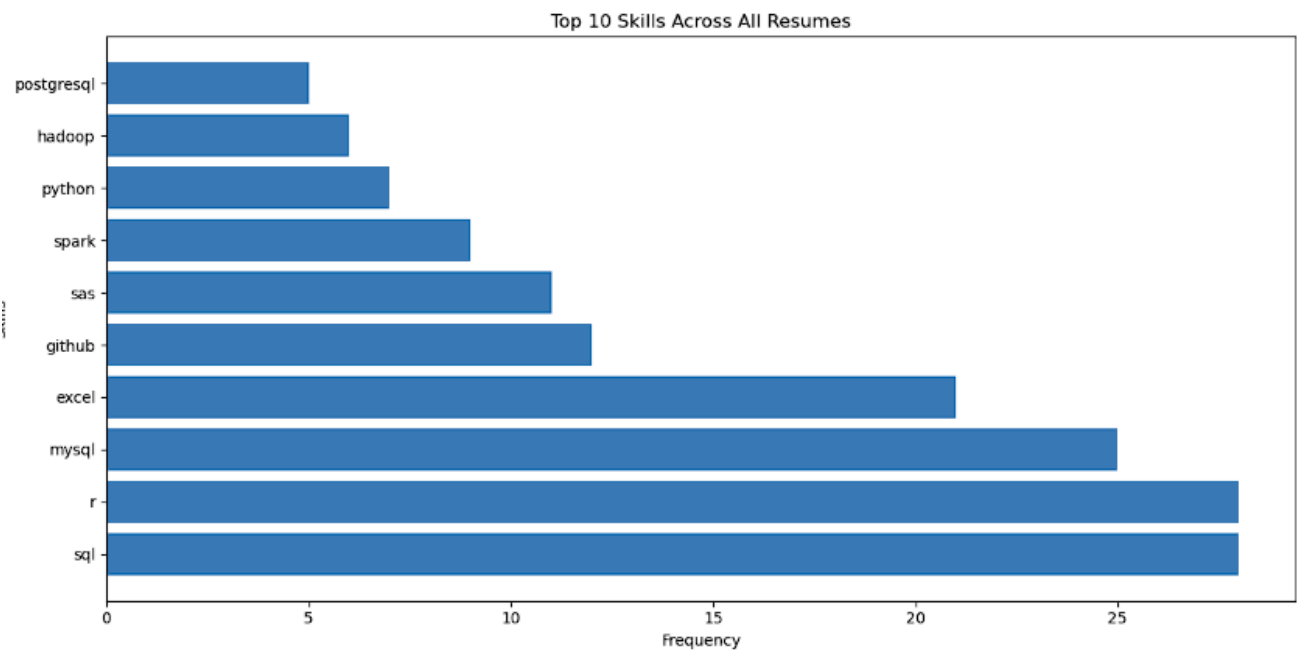
The system also used skill count metrics during ATS score generation and candidate evaluation processes.

5.4 TOP SKILLS FREQUENCY ANALYSIS

Skill frequency analysis was performed to identify the most common technical skills across all resumes within the dataset.

This analysis helps recruiters understand market trends, candidate technology preferences, and high-demand technical competencies.

Figure 5.4.1: Top Skills Across All Resumes



The graph represents the frequency distribution of technical skills identified across candidate resumes. Skills such as SQL, R Programming, MySQL, Excel, and GitHub were among the most frequently occurring technical competencies.

Skill frequency analysis can support recruitment planning, candidate filtering, and workforce analytics.

5.5 SKILL SCORE GENERATION

The ATS system generated skill scores for each candidate based on the number and relevance of technical skills identified within resumes.

Higher skill scores were assigned to candidates containing a larger number of relevant technical keywords and competencies related to the job description.

Figure 5.5.1: Skill Score Dataset

	resume_name	skills_found	skill_score
0	Abiral_Pandey_Fullstack_Java.pdf	[sql, excel, r, sas, mysql, postgresql, github]	30
1	Anudeep N_Sr Java Developer.pdf	[sql, r, sas, mysql, github]	20
2	Ashwini J2EE Developer.pdf	[sql, excel, r, mysql, postgresql, github, spark]	30
3	Bapuji Hadoop developer.pdf	[python, sql, excel, tableau, r, mysql, github...]	50
4	CHETAN_Sr Java Developer.pdf	[sql, excel, r, mysql]	20

The figure shows skill scores generated for different candidate profiles after evaluating extracted technical competencies.

Skill scores formed an important component of the final ATS matching percentage and candidate ranking calculations.

5.6 IMPORTANCE OF SKILL ANALYSIS IN RECRUITMENT

Technical skill analysis plays a critical role in modern recruitment systems because recruiters often prioritize candidates possessing relevant industry competencies.

Automated skill extraction systems improve recruitment efficiency by:

- Reducing manual resume analysis effort
- Identifying relevant technical candidates quickly
- Improving ATS score accuracy
- Supporting data-driven hiring decisions
- Enhancing candidate filtering operations
- Enabling recruitment analytics and workforce planning

Chapter 6: Candidate Matching and Ranking Model

6.1 INTRODUCTION TO CANDIDATE MATCHING SYSTEMS

Candidate matching is one of the most important functionalities of modern Applicant Tracking Systems (ATS). Recruitment organizations receive large numbers of resumes for every job opening, making manual candidate evaluation difficult and time-consuming.

AI-powered recruitment systems solve this problem by automatically comparing candidate resumes with job descriptions and identifying the most relevant applicants according to technical skills, keywords, experience, and textual similarity.

In this project, a machine learning-based candidate ranking model was developed using TF-IDF vectorization and cosine similarity algorithms. The system analyses resume content, calculates ATS matching percentages, and ranks candidates according to their relevance to the uploaded job description.

The implemented ranking model helps recruiters identify highly suitable candidates efficiently while reducing manual screening effort.

6.2 TF-IDF VECTORIZATION

TF-IDF (Term Frequency–Inverse Document Frequency) is one of the most widely used Natural Language Processing techniques for converting textual data into numerical feature vectors suitable for machine learning analysis.

TF-IDF measures the importance of words within textual documents by evaluating:

- Term Frequency (TF) – frequency of a word within a document
- Inverse Document Frequency (IDF) – uniqueness of a word across multiple documents

Words that appear frequently within a resume but less frequently across other resumes receive higher TF-IDF scores because they contain stronger distinguishing information.

In this project, TF-IDF vectorization was applied to cleaned resume text and job descriptions to generate machine-readable numerical representations suitable for similarity analysis. The generated TF-IDF vectors formed the foundation for candidate matching and ATS score calculations.

6.3 COSINE SIMILARITY ANALYSIS

After TF-IDF vectorization, cosine similarity algorithms were applied to compare candidate resumes against job descriptions.

Cosine similarity measures the angle between two numerical vectors and determines how similar the documents are in terms of textual content and technical keywords.

The similarity score ranges between 0 and 1:

- Higher similarity scores indicate stronger resume-job relevance
- Lower similarity scores indicate weaker candidate-job alignment

The ATS system used cosine similarity scores to calculate candidate matching percentages and rank resumes automatically.

Cosine similarity is highly effective for recruitment systems because it focuses on textual relevance rather than exact keyword positions.

6.4 CANDIDATE RANKING MODEL

The candidate ranking model combines:

- Skill Scores
- TF-IDF Similarity Scores
- ATS Matching Percentages
- Resume Quality Metrics

to generate final candidate rankings.

The system compares candidate resumes against uploaded job descriptions and automatically identifies the most suitable candidates based on technical relevance and skill alignment.

Candidates with higher similarity scores and stronger technical skill profiles receive higher ATS rankings.

Figure 6.1.1: Candidate Matching Dataset

	resume_name	skill_score	similarity_score	match_percentage
14	Niteesh Java Developer.pdf	40	0.146991	100.000000
3	Bapuji Hadoop developer.pdf	50	0.088774	95.176116
18	Sai Srinivas_Sr_Java_Developer.pdf	40	0.108027	87.763342
25	Venkata_SrPHP_Developer.pdf	30	0.145222	85.985031
12	Mani_Hadoop.pdf	40	0.093595	83.230969
17	RaviBurra_Certified PM_DevOps.pdf	30	0.123794	79.255553
24	Tarun_Developer.pdf	30	0.123542	79.176417
26	vikas java.pdf	30	0.026404	48.670268
23	Sundar_Java_8 Years.pdf	30	0.026404	48.670268
22	Sumanth Manne Java Developer.pdf	30	0.026404	48.670268

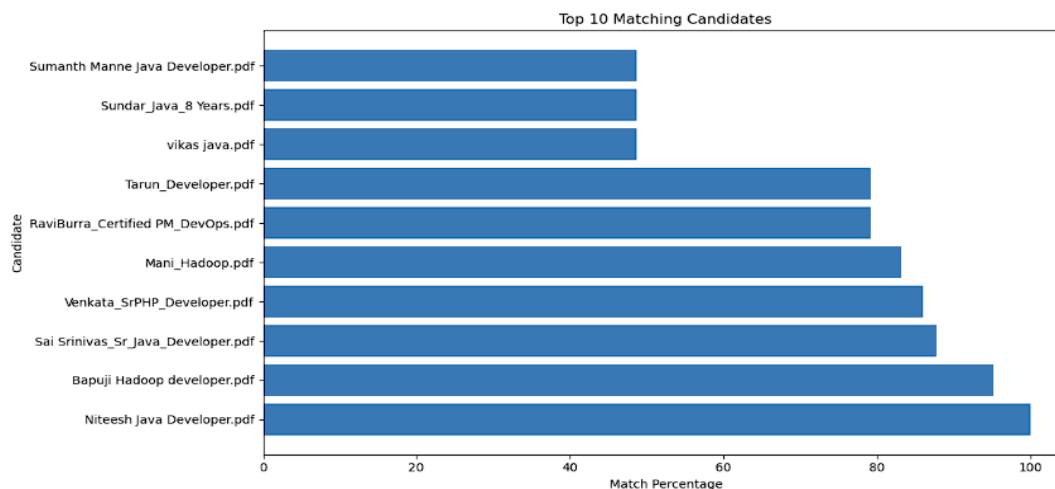
The above figure shows ATS matching percentages and similarity scores generated for candidate resumes after TF-IDF and cosine similarity analysis.

The ranking system successfully identified highly relevant candidates according to job description similarity and technical skill representation.

6.5 TOP CANDIDATE RANKING VISUALIZATION

Graphical candidate ranking analysis was performed to visualize ATS matching percentages and candidate performance rankings.

Figure 6.5.1: Top Matching Candidates



The graph represents the highest-ranked candidates identified by the ATS system based on similarity analysis and skill matching operations.

Candidates with higher ATS percentages were categorized as highly relevant applicants for the target job description.

The ranking visualization helps recruiters quickly identify suitable candidates without manually reviewing every resume individually.

6.6 ATS SCORE CALCULATION

The ATS system generated final ATS percentages by combining similarity scores and technical skill analysis.

The final ATS score represents the overall relevance of a candidate profile relative to the uploaded job description.

The scoring model considered:

- Technical Skill Presence
- Resume Similarity
- Keyword Relevance
- Resume Completeness
- Candidate Technical Diversity

Higher ATS percentages indicate stronger alignment between candidate profiles and job requirements.

The ATS scoring mechanism improved recruitment efficiency by enabling automated candidate shortlisting and ranking.

Chapter 7: ATS Dashboard and Recruitment Analytics System

7.1 INTRODUCTION TO RECRUITMENT DASHBOARDS

Modern recruitment systems require interactive dashboards capable of presenting analytical insights, candidate rankings, ATS scores, and recruitment metrics in a user-friendly manner. Recruiters often need quick access to candidate information, filtering systems, and visualization tools for efficient decision-making.

Traditional recruitment processes generally involve manually reviewing resumes and maintaining candidate information using spreadsheets or static databases. Such approaches become inefficient when organizations receive large volumes of applications for multiple job openings.

Interactive recruitment dashboards solve these challenges by integrating Artificial Intelligence, data visualization, and analytical reporting within a single platform.

In this project, an interactive ATS dashboard was developed using Streamlit to provide recruiters with an efficient interface for candidate analysis, filtering, ATS score evaluation, and recruitment analytics.

7.2 STREAMLIT-BASED ATS DASHBOARD

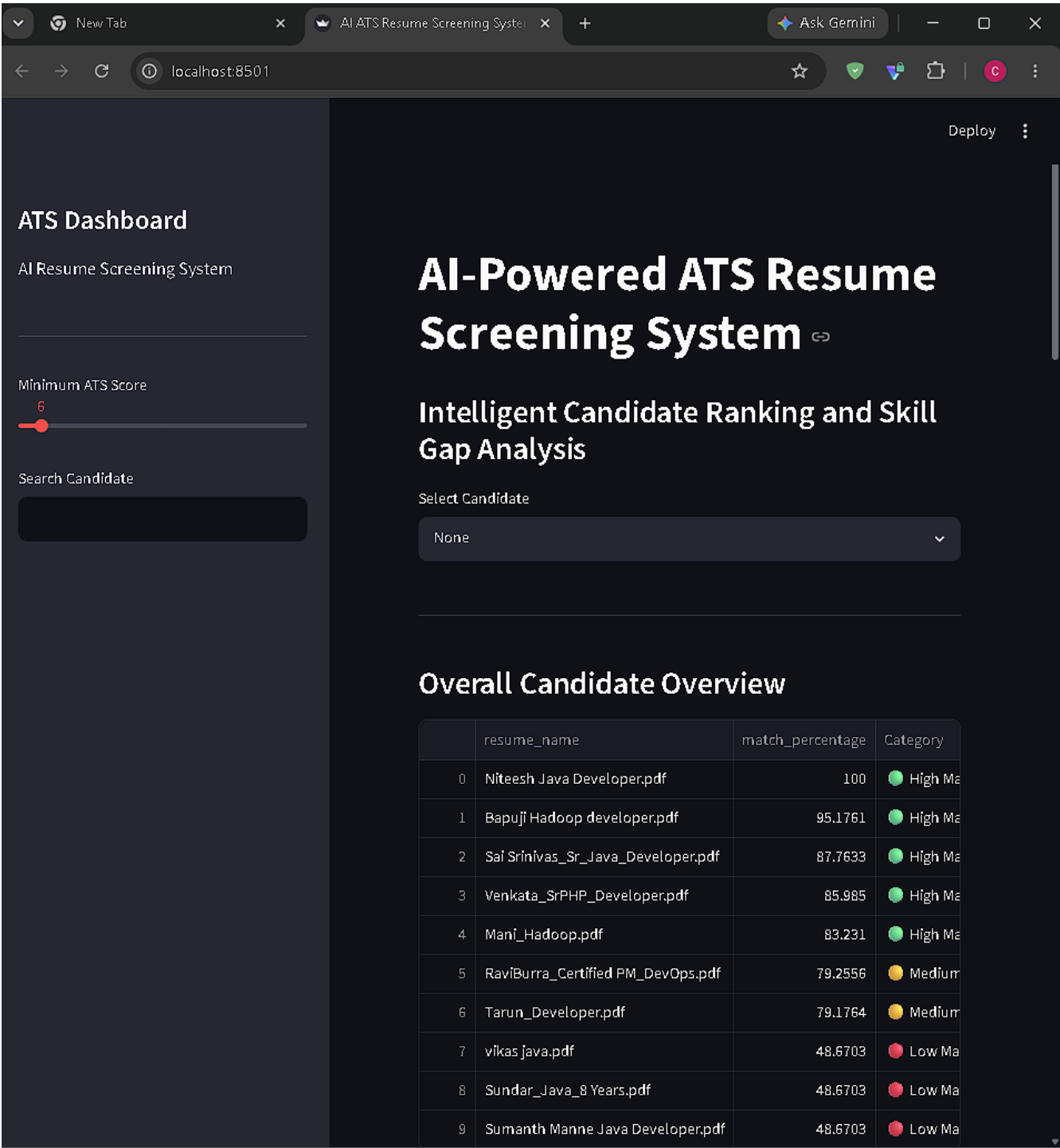
The ATS dashboard was developed using the Streamlit framework, which enables rapid deployment of machine learning and analytical applications using Python.

The dashboard provides a user-friendly recruiter interface supporting :-

- Candidate Filtering
- ATS Score Analysis
- Candidate Ranking Visualization
- Resume Matching Analysis
- CSV Export Functionality
- Skill Gap Analysis
- Interactive Candidate Selection

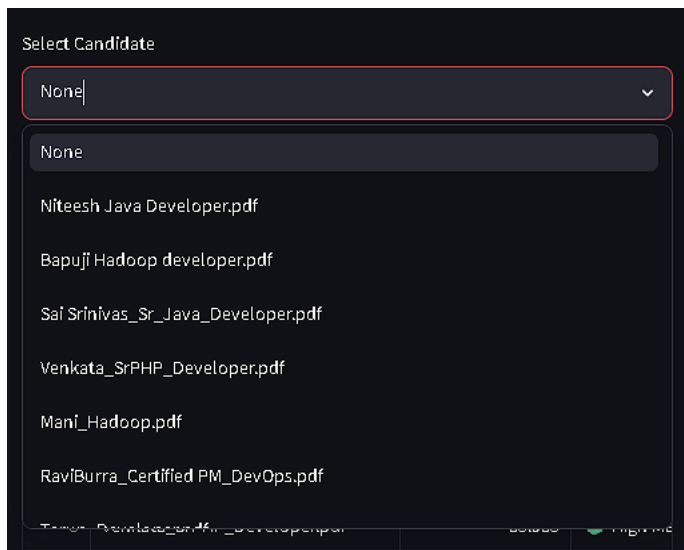
The dashboard was designed to simplify recruitment workflows and improve recruiter productivity through analytical automation.

Figure 7.2.1: ATS Dashboard Interface



The above figure shows the primary ATS dashboard developed using Streamlit. The interface displays candidate rankings, ATS percentages, filtering systems, and recruitment analytics within a centralized recruiter-oriented platform. The dark-themed interface improves readability while presenting analytical outputs in an organized format.

7.3 CANDIDATE FILTERING SYSTEM



The ATS platform includes an interactive filtering mechanism allowing recruiters to filter candidates according to ATS scores and candidate relevance.

Recruiters can:

- Filter candidates based on minimum ATS percentage
- Search candidate profiles
- Identify high-match candidates
- Shortlist suitable applicants quickly

This functionality improves recruitment efficiency by reducing manual candidate screening effort.

Figure 7.3.2: Candidate Filtering Dashboard

	ATS_Rank	resume_name	match_percentage
0	1	Niteesh Java Developer.pdf	100
1	2	Bapuji Hadoop developer.pdf	95.1761
2	3	Sai Srinivas_Sr_Java_Developer.pdf	87.7633
3	4	Venkata_SrPHP_Developer.pdf	85.985
4	5	Mani_Hadoop.pdf	83.231
5	6	RaviBurra_Certified PM_DevOps.pdf	79.2556

The dashboard enables recruiters to dynamically analyse candidate rankings and recruitment insights using interactive filtering components.

7.4 CANDIDATE RANKING TABLE

The dashboard displays structured candidate ranking tables containing ATS scores, candidate names, and recruitment categories.

Candidates were categorized into:

- High Match
- Medium Match
- Low Match

based on ATS percentage calculations and similarity analysis.

Figure 7.4.1: Candidate Ranking Table

	resume_name	match_percentage	Category
0	Niteesh Java Developer.pdf	100	 High Ma
1	Bapuji Hadoop developer.pdf	95.1761	 High Ma
2	Sai Srinivas_Sr_Java_Developer.pdf	87.7633	 High Ma
3	Venkata_SrPHP_Developer.pdf	85.985	 High Ma
4	Mani_Hadoop.pdf	83.231	 High Ma
5	RaviBurra_Certified PM_DevOps.pdf	79.2556	 Medium
6	Tarun_Developer.pdf	79.1764	 Medium
7	vikas java.pdf	48.6703	 Low Ma

The ranking table helps recruiters quickly identify highly suitable candidates according to ATS scores and technical relevance.

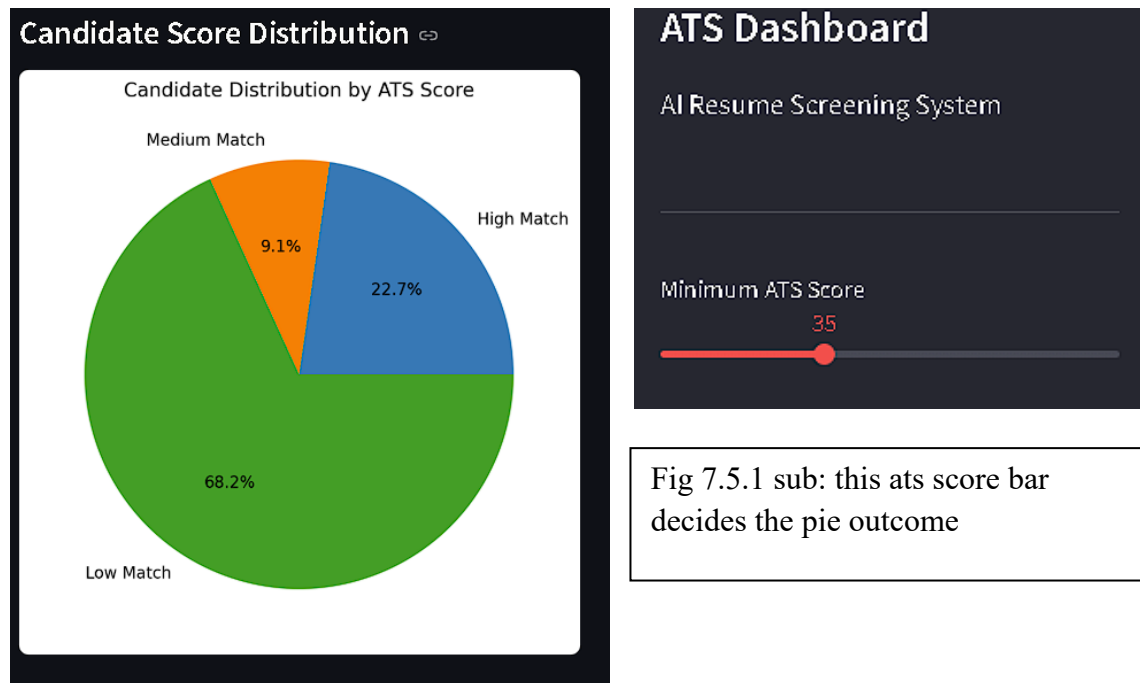
The categorization system improves decision-making efficiency during recruitment screening operations.

7.5 ATS SCORE DISTRIBUTION ANALYSIS

The ATS dashboard also includes graphical visualization systems for analysing candidate score distributions.

Visualization techniques help recruiters understand overall candidate quality distribution and recruitment trends.

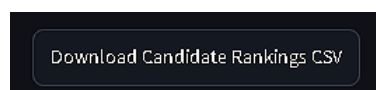
Figure 7.5.1: ATS Score Distribution Graph



The graphical visualization represents the distribution of candidates across different ATS score categories including Excellent, Good, Average, and Low Match groups.

This analytical representation helps recruiters evaluate the overall quality of candidate applications within the recruitment dataset.

7.6 CSV EXPORT FUNCTIONALITY



The ATS platform supports CSV export functionality allowing recruiters to download candidate ranking reports and ATS analysis results.

Export functionality improves usability by enabling:

- Recruitment reporting
- Candidate documentation
- Analytical sharing
- Recruitment record maintenance

The exported reports can be used for further HR analytics and recruitment management operations.

7.8 ADVANTAGES OF DASHBOARD-BASED ATS SYSTEMS

Dashboard-oriented recruitment systems provide several practical advantages in modern hiring environments:

- Improved recruiter efficiency
- Faster candidate analysis
- Better hiring transparency
- Simplified recruitment workflows
- Real-time analytical insights
- Automated reporting systems
- Improved candidate shortlisting accuracy

Interactive ATS dashboards significantly reduce manual effort while improving recruitment productivity and analytical accessibility.

7.9 SYSTEM LIMITATIONS AND FUTURE IMPROVEMENTS

Although the implemented ATS dashboard successfully demonstrates recruitment analytics and candidate ranking functionalities, certain limitations remain.

The current dashboard primarily focuses on textual resume analysis and does not include:

- Real-time cloud deployment
- Advanced deep learning models
- Multilingual resume analysis
- Interview scheduling integration
- Enterprise HR database connectivity

Chapter 8: Results and Insights

8.1 INTRODUCTION

The implementation of the AI-Powered ATS Resume Screening and Candidate Ranking System successfully demonstrated how Artificial Intelligence, Natural Language Processing (NLP), and machine learning techniques can automate recruitment workflows and improve candidate evaluation efficiency.

The system successfully processed resume data, extracted technical skills, calculated ATS matching percentages, generated candidate rankings, and provided recruiter-oriented analytical visualizations through an interactive dashboard interface.

Multiple analytical models and visualization systems were implemented to evaluate candidate quality, technical competencies, resume completeness, and job relevance.

8.2 RESUME ANALYSIS RESULTS

The preprocessing and resume parsing pipeline successfully extracted textual information from PDF resumes and converted unstructured resume content into machine-readable analytical datasets.

The system successfully performed:

- Resume text extraction
- Resume length analysis
- Resume completeness analysis
- NLP preprocessing
- Technical skill identification

The preprocessing stage improved text quality and reduced unnecessary information before machine learning analysis.

Key Observation:

Resumes containing detailed technical skills, projects, and certifications generally produced better ATS scores and stronger similarity rankings.

8.3 SKILL EXTRACTION RESULTS

The skill extraction model successfully identified technical skills from candidate resumes using NLP preprocessing and keyword analysis techniques.

The system detected important technologies including:

SQL	Python	R Programming	MySQL
Hadoop	GitHub	Tableau	Spark
Excel	Java		

The extracted skills were used for ATS score generation and candidate ranking operations.

8.4 CANDIDATE RANKING RESULTS

The TF-IDF and cosine similarity model successfully compared resumes against uploaded job descriptions and generated ATS matching percentages.

Candidates were automatically ranked according to:

- Technical skill relevance
- Keyword similarity
- Resume completeness
- ATS score calculations

Key Observation:

Candidates with stronger technical keyword alignment and detailed project descriptions consistently achieved higher similarity scores.

8.5 ATS SCORE ANALYSIS

The ATS scoring system successfully categorized candidates into multiple recruitment categories based on their matching percentages.

The categories included:

- Excellent Match
- Good Match
- Average Match
- Low Match

8.6 DASHBOARD AND VISUALIZATION RESULTS

The Streamlit dashboard successfully integrated all analytical modules into a centralized recruiter-oriented recruitment platform.

The dashboard supported:

- Interactive candidate filtering
 - ATS score visualization
 - Ranking analysis
 - Recruitment analytics
 - CSV export functionality
 - Candidate categorization
-

8.7 OVERALL SYSTEM PERFORMANCE

The implemented ATS Resume Screening System successfully demonstrated the practical application of Artificial Intelligence and Natural Language Processing in recruitment automation.

The project achieved:

- Automated resume screening
- Intelligent candidate ranking
- Skill extraction automation
- ATS score generation
- Recruitment visualization analytics
- Improved recruiter productivity

The integration of NLP preprocessing, TF-IDF vectorization, cosine similarity analysis, and dashboard visualization created a complete AI-powered recruitment analytics platform.

Chapter 9: Conclusion and Future Scope

9.1 CONCLUSION

The AI-Powered ATS Resume Screening and Candidate Ranking System successfully demonstrated how Artificial Intelligence, Natural Language Processing (NLP), and machine learning techniques can be applied to modern recruitment automation systems.

The project addressed one of the major challenges faced by recruiters and organizations — the manual screening and evaluation of large numbers of candidate resumes.

Traditional recruitment methods often require substantial time, effort, and human resources, making the hiring process inefficient and inconsistent. The implemented ATS platform successfully automated multiple stages of the recruitment workflow including resume parsing, text preprocessing, skill extraction, ATS score generation, candidate ranking, and recruitment analytics visualization.

The system integrated several technologies including Python, Pandas, Streamlit, SpaCy, TF-IDF vectorization, cosine similarity analysis, and visualization libraries to create an intelligent recruitment assistance platform. Resume data was extracted from PDF documents, processed using NLP preprocessing techniques, and analysed using machine learning-based similarity algorithms.

Five independent analytical models were successfully implemented within the system:

1. Resume Length Analysis Model
2. Resume Completeness Analysis Model
3. Skill Extraction and Frequency Analysis Model
4. Candidate Matching and Ranking Model
5. ATS Dashboard and Recruitment Analytics Model

These models collectively improved recruitment automation efficiency and enabled intelligent candidate evaluation processes.

The skill extraction module successfully identified important technical competencies from resumes, while TF-IDF and cosine similarity algorithms effectively compared resumes against job descriptions to generate ATS matching percentages and candidate rankings. The Streamlit-based dashboard provided recruiters with interactive analytical visualizations, candidate filtering systems, ATS score analysis, and recruitment insights.

Overall, the project established a strong foundation for intelligent recruitment systems capable of improving hiring productivity, recruiter efficiency, and candidate evaluation consistency.

9.2 FUTURE SCOPE

Although the implemented ATS Resume Screening System successfully demonstrated intelligent recruitment automation, several improvements and advanced functionalities can be incorporated in future versions of the platform.

Future enhancements may include:

- **Deep Learning-Based Resume Analysis**

Advanced neural network and transformer-based NLP models such as BERT and GPT can be integrated for semantic resume understanding and contextual candidate analysis.

- **Multilingual Resume Processing**

Future systems may support resumes written in multiple languages for global recruitment applications.

- **Cloud-Based Deployment**

The ATS platform can be deployed on cloud infrastructure to support large-scale enterprise recruitment operations and real-time resume processing.

- **Interview Recommendation Systems**

AI-driven interview scheduling and candidate recommendation modules can be integrated into the platform.

- **Real-Time Recruitment Database Integration**

Future systems may connect directly with HR databases, LinkedIn profiles, and online job portals for automated candidate management.

- **Advanced Semantic Skill Matching**

Semantic similarity algorithms can improve recruitment accuracy by identifying conceptually related skills rather than relying only on keyword-based matching.

- **AI Chatbot Integration**

Recruitment chatbots can be integrated to assist recruiters and candidates during hiring workflows.

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13. Scikit-learn Documentation. Available at: <https://scikit-learn.org/>
14. Pandas Documentation. Available at: <https://pandas.pydata.org/>
15. NumPy Documentation. Available at: <https://numpy.org/>
16. Python Software Foundation. Python Documentation. Available at: <https://www.python.org/>
17. Research articles and technical resources related to Artificial Intelligence, Applicant Tracking Systems, Natural Language Processing, and Recruitment Analytics.

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